END-TO-END MACHINE LEARNING PROJECT

Predicting California House Prices

From a neural network to a running app: tracked in MLflow, served with FastAPI, packaged in Docker, used through Streamlit.

TensorFlow · MLflow · FastAPI · Docker · Streamlit



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01 · THE DATA

20,640 districts, 13 features

20,640

rows in housing.csv

California districts, 1990 census

13

model features

8 numeric + 5 one-hot ocean_proximity

207

missing values

total_bedrooms, filled with the median

80 / 20

train / test split

16,512 train rows, 4,128 test rows

PREPARATION STEPS

1. **Load** housing.csv
2. **Fill missing** bedrooms
3. **One-hot** ocean_proximity
4. **Train / test** split
5. **StandardScaler** (scaler.pkl)

TARGET median_house_value, from about $15,000 to $500,000. The scaler is fitted on training data only and reused by the API.

02 · THE MODELS

Three neural networks, one fair test

Baseline MLP

The simple reference

- Dense 64
- Dense 32
- Dense 1 (price)

Deeper DNN

More layers, more capacity

- Dense 128
- Dense 64
- Dense 32
- Dense 16
- Dense 1 (price)

Regularised DNN

Fights overfitting

- Dense 64
- BatchNorm
- Dropout 0.1
- Dense 32
- BatchNorm
- Dropout 0.2
- Dense 1 (price)

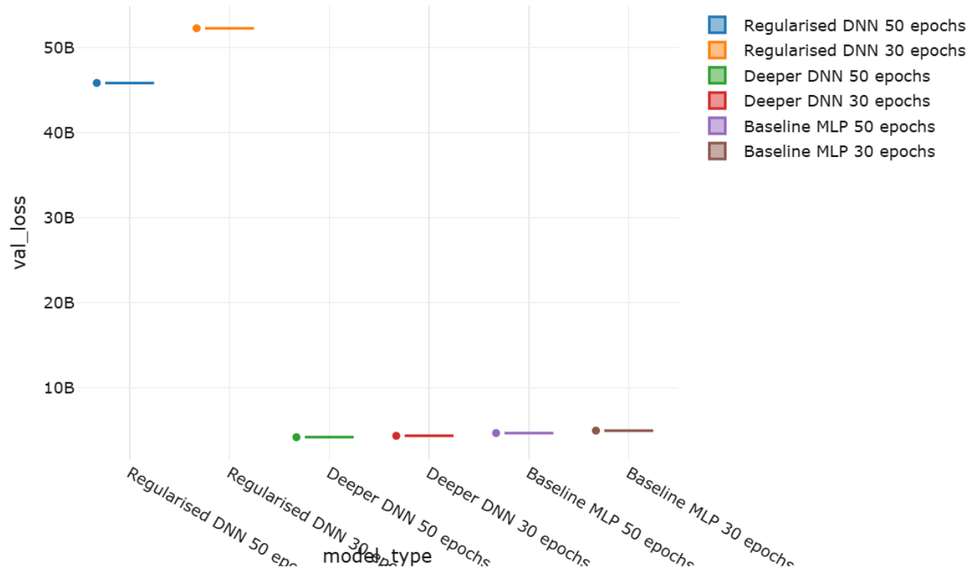
SAME SETUP FOR ALL Adam · MSE loss · MAE metric · batch size 30 · 20% validation split · 30 vs 50 epochs = 6 MLflow runs

03 · RESULTS

Six runs compared in MLflow

MODEL	EPOCHS	VAL LOSS	VAL MAE
Deeper DNN	50	**4.18B**	$44,919
Deeper DNN	30	**4.34B**	$46,068
Baseline MLP	50	**4.66B**	$48,322
Baseline MLP	30	**4.95B**	$49,817
Regularised DNN	50	**45.8B**	$188,360
Regularised DNN	30	**52.3B**	$200,298

Lower is better. Best run highlighted.



MLflow chart: validation loss for each run

Both regularised runs sit near 46B to 52B; the rest are near 4B to 5B.

04 · THE WINNER

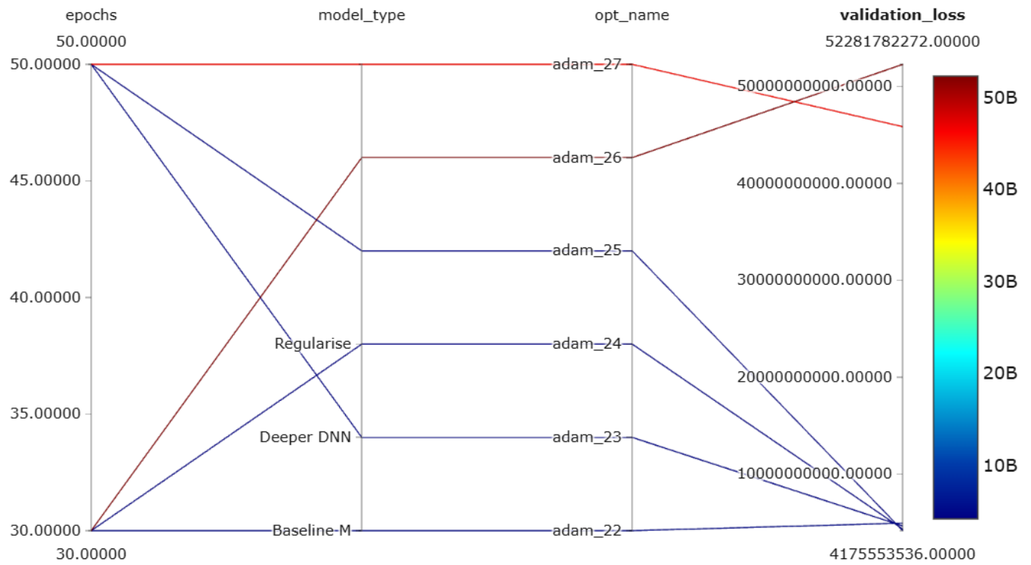
Deeper DNN, 50 epochs

4.18B
validation loss

$44,919
validation MAE

$64.6K
RMSE, from the loss

- Lowest validation loss of all six runs, and about 10% lower than the best baseline (4.66B).
- Training loss 3.85B vs validation loss 4.18B: close together, so little overfitting.
- The regularised runs were roughly 10x worse (46B to 52B).
- Picked automatically: runs sorted by val_loss in MLflow, saved as best_model.keras.



MLflow parallel coordinates plot
Lines that end lowest on validation_loss are the best runs.

05 · DEPLOYMENT

Two containers, one command



BUILD TIME

MLflow picks the best run. best_model.keras and scaler.pkl are copied into the backend image.

NETWORKING

Compose puts both services on one network, so Streamlit calls http://backend:8000/predict.

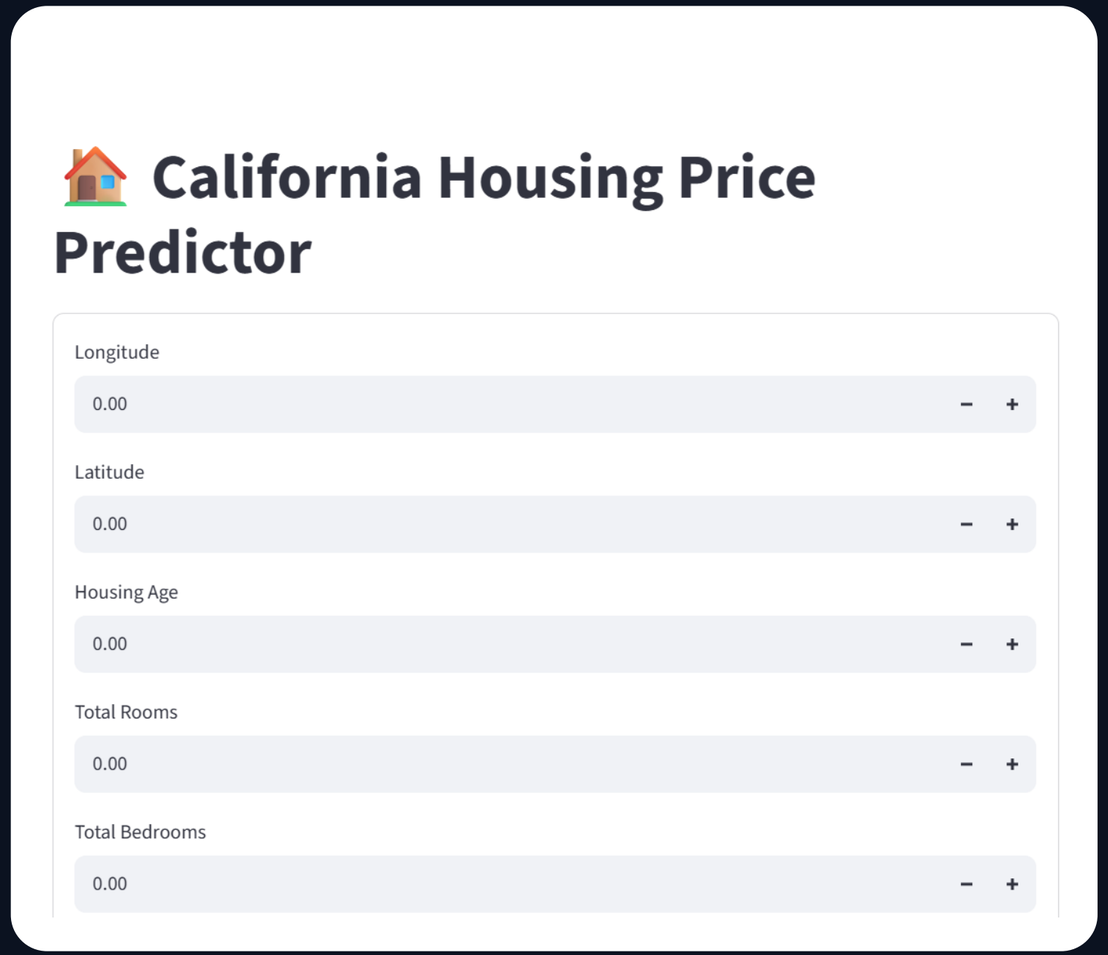
RUN IT

```
docker compose up --build -d
```

API docs localhost:8000/docs
App localhost:8501

06 · THE APP

Streamlit frontend, live predictions



1 Enter a house
Longitude, latitude, age, rooms, bedrooms, population, households, income and ocean proximity.

2 Send to the API
Streamlit posts the values as JSON to POST /predict.

3 Predict
FastAPI one-hot encodes, scales with scaler.pkl and runs the Deeper DNN.

4 Show the price
The estimated price comes back and is shown in a green box.

ENDPOINTS GET / health check POST /predict price /docs Swagger UI

07 · WRAP UP

What we built and what comes next

WHAT WE BUILT

- Cleaned data and trained 3 model types, 6 runs
- Tracked every run in MLflow and chose the best
- FastAPI service that serves the winning model
- Streamlit app for anyone to try
- Docker images and compose file: runs the same everywhere

NEXT STEPS

Scale the target price

Raw prices make the loss huge. Scaling should steady training.

Tune and stop early

Learning rate, batch size and early stopping.

Monitor and retrain

Drift monitoring and retraining jobs, as in our architecture plan.

Ship it

MLflow Model Registry, CI/CD and a cloud deployment.

Thank you. Questions?